Çekicin Vehicle AI v3 — what is new, why it is better, and the evidence

Plate reading (M7) · Make/model (M8) · Damage street test (M1) · 27 October 2026 · DEMO

Plate	Brand / model	Damage false alarms	Photos tested
92.8%	**86.6% / 55.2%**	**~10%**	**437**
250	67	250	+1000

🔗 https://cekicin-site.elasa2next.workers.dev/ai-lab/vehicle-ai.html · https://huggingface.co/sosa123454321/cekicin-plate-tr-onnx

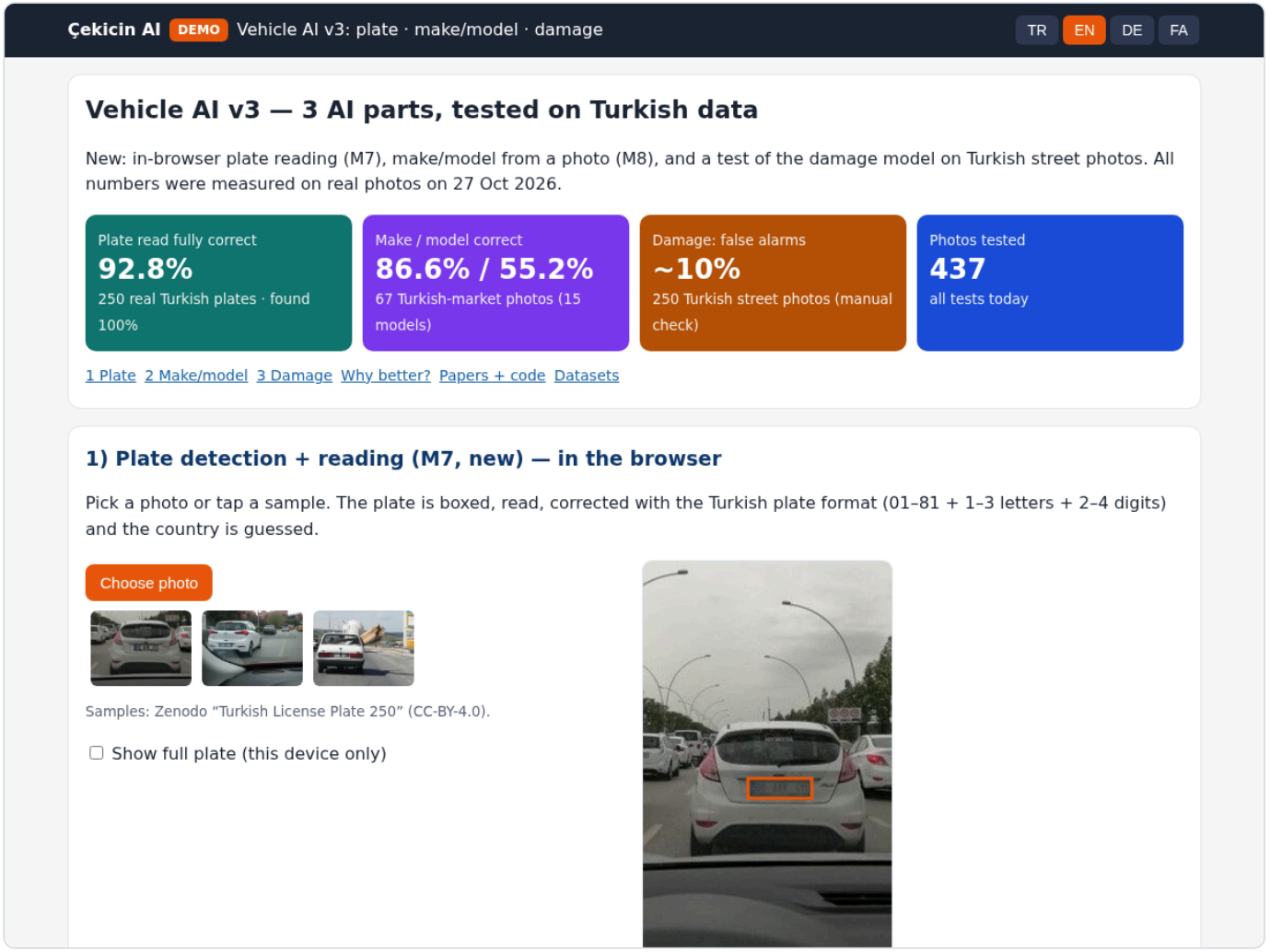
1. Summary

- **Plate (new):** 92.8% of 250 real Turkish plates read fully correct, 100% found, 98.4% characters; same model in the browser: 93.2%.
- **Make/model (new):** brand 86.6% / model 55.2% on 67 Turkish-market photos (15 models); brand 79.2% / model 50.8% on 120 Stanford Cars photos.
- **Damage:** first real Turkish street test: 250 photos, 35 flags → 6 real, 4 possible, 25 false (~10% of photos). Lab test: 1,000 photos, F1 0.59.
- **Photos tested today:** 437 (250 + 120 + 67); plus the earlier 1,000-photo damage test.
- **Papers with Code:** yes, found. The code of 15 papers/repos was checked; 3 MIT repos are already inside M7, 5 MIT/Apache repos are training candidates, 6 have no licence (read only).

3. Why is it better? (before → now)

Part	Before	Now (v3)	Evidence (photos)	Why better
Plate (M7)	none	**92.8% exact · 98.4% chars**	250 real Turkish plates (Zenodo TR-250); browser test: 93.2%	Runs on the device, photo never uploaded (KVKK). Note: the Turkish-format fixer changed only 1/250 results (not significant). Weak point: 22.8% on blurred photos
Make/model (M8)	none (HF dima806 covers 7 of the top-25 TR models)	**brand 86.6% · model 55.2%**	67 Turkish-market photos (15 models) + 120 Stanford Cars (brand 79.2%)	Knows Turkish cars (Tofaş Şahin, Egea, Duster); no training needed; Llama-4-Scout more stable than Mistral
Damage (M1)	lab test only (1,000 photos, F1 0.59)	**+ Turkish street test: 6 of 35 flags real**	250 Turkish street photos (manually checked) + 1,000 test photos	Measured the real-world weakness (windscreen photos); next version adds “no damage” negatives

2. What is new — screenshots



Top of the new page (4 languages: TR/EN/DE/FA). https://cekicin-site.elasa2next.workers.dev/ai-lab/vehicle-ai.html

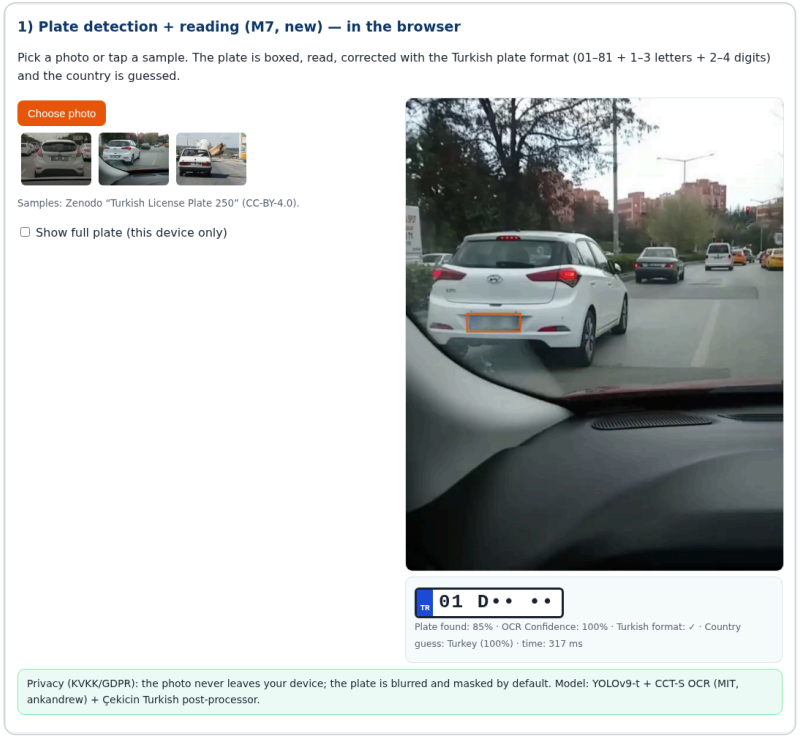
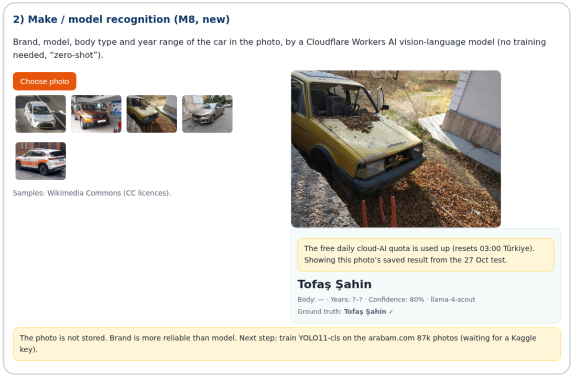


Plate demo: box + blur + masked text (KVKK). Full text only after the user ticks the box.



Make/model demo (live today: daily free AI quota used up by our tests → page shows the saved test result, clearly labelled).

4. Test details

4a. Plates — 250 real Turkish photos (Zenodo TR-250, CC-BY-4.0)

Detected 100% · exact (raw OCR) 92.4% → with Turkish format fixer 92.8% · characters 98.4% · 64 ms/photo on 2 CPUs (1.78 h per 100,000 photos). Browser (WASM, 1 thread): 93.2%, 219 ms.

Typical errors (truth → read): 06AFM986→06AFK986 · 06BBJ510→06GBJ510 · 06BG198→06BGR198 · 06BIS629→06BJS629 · 06DF2090→06BF2090 · 06DIF47→06DFF47 · 06DP351→06DP851 · 06DTH92→06DTG92 · 06FH9483→06FJ9483 · 06GUD11→06GU01

4b. Make/model — Turkish market photos (Wikimedia Commons, unique photos)

Model	Photos	Brand ✓	Model ✓	Wrong answers
Renault Clio	5	5/5	3/5	Renault Captur, Renault Megane
Renault Megane	2	2/2	2/2	
Tofaş Şahin	4	4/4	4/4	
Togg T10X	4	0/4	0/4	BYD Atto 3, Fiat Egea, Geely Kx11, Skylark X55
Dacia Duster	6	6/6	6/6	
Fiat Linea	6	5/6	3/6	Fiat Egea, Renault Fluence
Hyundai i20	6	6/6	2/6	Hyundai Kona, Hyundai i30
Renault Taliant	3	3/3	0/3	Renault Megane, Renault Talisman
Opel Astra	4	4/4	4/4	
Volkswagen Passat	4	3/4	2/4	Skoda Octavia, Volkswagen Jetta
Peugeot 301	6	6/6	3/6	Peugeot 308, Peugeot 408
Citroën C-Elysée	5	3/5	0/5	Chery QQ, Citroen C4, Fiat Egea
Fiat Egea	1	1/1	1/1	
Toyota Corolla	6	6/6	6/6	
Renault Symbol	5	4/5	1/5	Fiat Egea, Renault Clio, Renault Fluence, Renault Thalia
Σ	67	**86.6%**	**55.2%**	

Model comparison: Llama-4-Scout brand 86.6% (0 errors) vs Mistral-Small-3.1 brand 75.0% on the 28 photos it answered — Mistral was stopped by the daily quota (429), so Llama-4-Scout was chosen. Weak points: Togg T10X (too new for the model), Renault Taliant and Citroën C-Elysée (brand right, model wrong).

4c. Damage — Turkish street photos

The current damage model (YOLO11m, on-device) was run on the 250 TR-250 street photos and every flag was checked by hand: 35 flags = 6 real damage, 4 possible, 25 false alarms. Most false alarms are dashboard/windscreen reflections. Fix for v4: add “no damage” Turkish street photos as negatives and raise thresholds for glass/lamp classes. https://cekicin-site.elasa2next.workers.dev/ai-lab/model-test.html

5. Comparison with academic papers

Papers with Code (paperswithcode.com) was shut down on 24 July 2025; we used its official frozen archive (HF pwc-archive, July 2025). Papers use their own benchmarks, so numbers are indicative, not head-to-head. No leaderboard exists for car damage.

Part	Paper	Data	Result	Ours
Plate	Laroca et al. 2021 — Efficient & Layout-Independent ALPR (YOLO + CR-NET) https://arxiv.org/abs/1909.01754	AOLP · OpenALPR-EU · UFPR-ALPR	99.2 · 97.8 · 90.0 %	TR-250: 92.8%
Plate	Character Time-series Matching (2023) https://arxiv.org/abs/2307.11336	UFPR-ALPR	96.7 %	TR-250: 92.8%
Plate	Silva & Jung, ECCV 2018 — Unconstrained ALPR (WPOD-NET) http://openaccess.thecvf.com/content_ECCV_2018/html/Sergio_Silva_License_Plate_Detection_ECCV_2018_paper.html	AOLP-RP	98.36 %	—
Plate	LPRNet (2018) — real-time plate OCR http://arxiv.org/abs/1806.10447	Chinese plates	0.34 GFLOPs	13 MB, 64 ms CPU
Make/model	Liu et al. 2024 — Progressive Multi-task Anti-Noise Learning (PMD/PMAL) https://arxiv.org/abs/2401.14336	Stanford Cars · CompCars	97.3 · 99.1 %	Stanford-120 brand 79.2% / model 50.8% (zero-shot)
Make/model	CMAL-Net (Pattern Recognition 2023) https://www.sciencedirect.com/science/article/pii/S0031320323002509	Stanford Cars	97.1 %	—
Make/model	I2-HOFI (IJCV 2024) https://link.springer.com/article/10.1007/s11263-024-02260-y	Stanford Cars	96.92 %	—
Make/model	EfficientNetV2 (2021) https://arxiv.org/abs/2104.00298	Stanford Cars	95.1 %	—
Damage	CarDD (IEEE T-ITS 2023) — Car Damage Detection dataset + DCN+ https://arxiv.org/abs/2211.00945	CarDD	box AP 60.6 · mask AP 57.0	CarDD AP 62.3 · AP50 78.1
Damage	RF-DETR (2025) — real-time detection transformer https://arxiv.org/abs/2511.09554	COCO / RF100-VL	SOTA real-time	next-step candidate

6. Papers with Code — did we find code, and can we add it?

Yes. Every listed paper has GitHub code. **Added now:** fast-plate-ocr + open-image-models (MIT) run inside the new plate reader (M7), with our Turkish post-processing. **Can be added next:** TITS-LPST, anti-noise_fgvr, timm, ImageNet21K, RF-DETR (MIT/Apache). **Cannot be copied:** repos without a licence (read-only); Ultralytics is AGPL (needs open-sourcing or a commercial licence).

Part	Code	Licence	★	Status
Plate	github.com/ankandrew/fast-plate-ocr OCR model used in M7	MIT	797	added
Plate	github.com/ankandrew/open-image-models YOLOv9-t plate detector used in M7	MIT	123	added
Plate	github.com/ankandrew/fast-alpr pipeline reference	MIT	844	added
Plate	github.com/brightyoun/TITS-LPST arXiv 1909.01754	MIT	40	candidate
Plate	github.com/sergiomsilva/alpr-unconstrained ECCV 2018	NOASSERTION	1771	reference only

Plate	github.com/chequanghuy/Character-Time-series-Matching arXiv 2307.11336	—	57	reference only
Plate	github.com/lyl8213/Plate_Recognition-LPRnet arXiv 1806.10447	—	285	reference only
Make/model	github.com/dichao-liu/anti-noise_fgvr arXiv 2401.14336 (97.3 / 99.1 %)	Apache-2.0	20	candidate
Make/model	github.com/huggingface/pytorch-image-models EfficientNetV2 / TResNet backbones	Apache-2.0	35k+	candidate
Make/model	github.com/Alibaba-MIIL/ImageNet21K TResNet-L 96.32 %	MIT	700+	candidate
Make/model	github.com/Dichao-Liu/CMAL CMAL-Net 97.1 %	—	48	reference only
Make/model	github.com/arindam-1991/i2-hofi I2-HOFI 96.92 %	—	5	reference only
Damage	github.com/roboflow/rf-detr arXiv 2511.09554	Apache-2.0	3k+	candidate
Damage	github.com/ultralytics/ultralytics YOLO11 (M1, AGPL: open-source our model or buy licence)	AGPL-3.0	40k+	added
Damage	github.com/CarDD-USTC/CarDD-USTC.github.io CarDD dataset: non-commercial	—	—	reference only

7. Datasets and photo counts

Part	Dataset	Size	Turkish?	Licence
Plate	Zenodo Turkish License Plate 250	250	yes	CC-BY-4.0
Plate	Kaggle synthetic-turkish-license-plates	100,000	yes	see Kaggle
Plate	Roboflow plakatanima / kemalkilicaslan	2,246 + 3,501	yes	CC-BY (Roboflow)
Make/model	Kaggle arabam.com car images	87,163 · 20 brands · 86 models	yes	Unknown
Make/model	Stanford Cars	16,185 · 196 classes	no	research
Make/model	Wikimedia Commons TR-market set (ours)	67 · 15 models	yes	CC (attributed)
Damage	VehiDE	13,945	no	CC-BY
Damage	CarDD	4,000	no	non-commercial
Damage	HF autodamageiq (Turkish labels)	15,183	yes	check (CarDD-derived)

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All results and scripts: in the HF repos and the report PDFs.

8. Limits and next steps

- Free Cloudflare AI quota (10,000 neurons/day) is small: ~300 car-photo checks per day. Our tests used it up today; resets 03:00 Türkiye time.
- Make/model is zero-shot: model name is right only ~55% of the time. Next: train YOLO11-cls / EfficientNetV2 on the Turkish arabam.com set (87,163 photos, 86 models) — needs a Kaggle API key from you.
- Licences: arabam.com set licence “unknown” (research only); CarDD non-commercial; plates are personal data (KVKK).
- Next test target: 100,000 photos (plan in the 46th report); plate speed 1.8 h per 100k photos on CPU.

9. Links

- **Vehicle AI v3:** https://cekicin-site.elasa2next.workers.dev/ai-lab/vehicle-ai.html
- **HF mirror:** https://sosa123454321-cekicin-site.static.hf.space/ai-lab/vehicle-ai.html
- **API /v1/vehicle/identify:** https://cekicin-ai-api.elasa2next.workers.dev/v1
- **HF model: cekicin-plate-tr-onnx:** https://huggingface.co/sosa123454321/cekicin-plate-tr-onnx
- **Damage test (model-test):** https://cekicin-site.elasa2next.workers.dev/ai-lab/model-test.html
- **Start hub:** https://cekicin-site.elasa2next.workers.dev/ai-lab/start/
- **Admin · Reports:** https://cekicin-site.elasa2next.workers.dev/admin/leads.html#project